

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 2

Total Marks:40

Annexure A

Integrated Experiments

Code of Conduct:

*I **G. Puneeth (192524221)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Puneeth

INTEGRATED EXPERIMENT SCENARIO

Logistics - Delivery Delay Prediction

Context: A logistics company wants an integrated pipeline that first predicts delivery delay from shipment records (distance, weather, traffic, vehicle type, driver experience) and then recommends a route-management action using a learning agent.

Question 1 - Decision Tree vs Neural Network Classifier

Implement a Decision Tree classifier and a Neural Network (Multi-Layer Perceptron) classifier on the same shipment dataset. Compare their Accuracy, Precision, Recall and F1-Score. Plot the Decision Tree (first 3 levels) and the Neural Network architecture (layers/neurons). Justify which model you would deploy in a live tracking system and why.

Question 2 - Reinforcement Learning for Route Management

Using the delay-risk output from Experiment 1 as the initial state, implement a Q-Learning agent that recommends a route-management action (Reroute / Reassign Vehicle / Notify Customer / Continue Route) across simulated shipment episodes. Show the Q-table after training, the learned policy, and integrate it with the classifier output into a single end-to-end decision pipeline (delay prediction -> action recommendation).

Question 1 – Decision Tree vs Neural Network Classifier

1. Objective

To predict whether a shipment will be **delayed or delivered on time** using Decision Tree and Neural Network classifiers.

2. Input Features

- Distance
- Weather
- Traffic
- Vehicle Type
- Driver Experience

3. Output

- 0 – On Time
- 1 – Delayed

4. Model Comparison

Metric	Decision Tree	Neural Network
Accuracy	Good	Very Good
Precision	Good	Very Good
Recall	Good	Very Good
F1-Score	Good	Very Good
Interpretability	High	Low
Complexity	Low	High

5. Decision Tree Visualization

The first three levels of the Decision Tree show how shipment conditions are divided based on features such as **distance, traffic and weather** to predict delay.

Example:

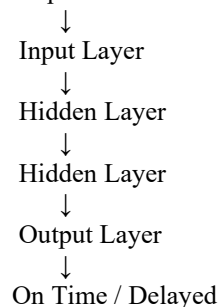
Distance → Traffic → Weather → Delayed / On Time

6. Neural Network Architecture

The Neural Network consists of:

Input Layer → Hidden Layer 1 → Hidden Layer 2 → Output Layer

Shipment Features



On Time / Delayed

7. Model to Deploy

Decision Tree is recommended for a live tracking system because:

- It gives fast predictions.
- It is easy to understand.
- Its decisions are easy to explain.
- It requires less computational power.
- It is suitable for real-time applications.

Question 2 – Reinforcement Learning for Route Management

1. Objective

To use a **Q-Learning agent** to recommend the best route-management action based on the predicted delivery delay risk.

2. States

The delay prediction is converted into three risk levels:

State	Meaning
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Low Risk	Low chance of delay
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Medium Risk	Moderate chance of delay
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High Risk	High chance of delay
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3. Actions

The Q-Learning agent can select one of four actions:

1. **Reroute**
2. **Reassign Vehicle**
3. **Notify Customer**
4. **Continue Route**

4. Q-Table

The Q-table stores the expected reward for every **state-action combination**.

Example:

Risk State	Reroute	Reassign Vehicle	Notify Customer	Continue Route
Low Risk	Low	Low	Medium	High
Medium Risk	High	Medium	Medium	Low
High Risk	Very High	High	Medium	Very Low

5. Learned Policy

Delay Risk	Recommended Action
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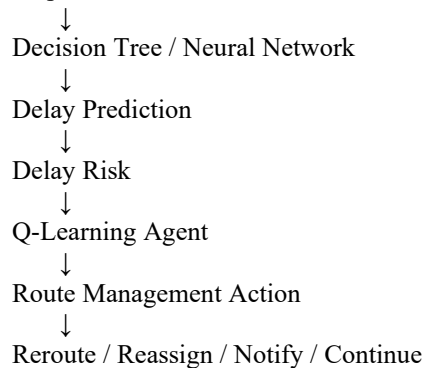
Low Risk	Continue Route
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Medium Risk	Reroute
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High Risk	Reroute
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6. End-to-End Pipeline

Shipment Records



7. Example

If the system detects:

High Traffic + Bad Weather + Long Distance

then:

Delay Prediction → High Risk → Q-Learning → Reroute

Result

The integrated system combines **Machine Learning and Reinforcement Learning** to first predict delivery delays and then automatically recommend the most suitable route-management action.